

Unit 1: Language Acquisition

Building Blocks

Phonemes (smallest distinguishable units) · **Morphemes** (smallest meaningful units; "disliked" = dis+lik+ed) · **Syntax** (rules for arrangement) · **Semantics** (meaning) · **Pragmatics** (use in context). Language = **generative** (infinite messages from finite symbols) + **rule-governed**.

Stages of Acquisition

Stage	Age	Features
Cooing	~2 mo	Vowel sounds; universal (incl. deaf)
Babbling	~6 mo	Consonant+vowel; shifts to own language phonemes (Locke, 1994); deaf infants stop
Holophrases	~12 mo	One-word utterances; usually nouns; overextension errors (e.g., "doggie" for any 4-legged animal)
Telegraphic speech	1.5–2.5 yr	Two-word; articles/prepositions omitted; e.g., "want juice," "doggie bite"
Adult structure	~4 yr	Complex constructions. Vocab: ~300 at 2, ~1000 at 3, adult-like by 10

4 Theories of Acquisition

Theory	Theorist	Central Idea
Behaviourist	Skinner (1957)	Imitation + reinforcement
Innateness	Chomsky (1957/59)	LAD; innate language faculty
Cognitive	Piaget	Language part of cognitive development
Interactionist	Bruner (1966, 1968)	CDS, scaffolding, LASS

Chomsky's Innateness Theory

LAD (Language Acquisition Device) = innate faculty. **Universal Grammar + Principles & Parameters**. Slobin: LAD = process mechanism. **Critical period** (Lenneberg, 1967): guaranteed up to 6, compromised until after puberty, rare thereafter. **Genie case** (1970, age 13): isolated, never acquired grammar. **Williams Syndrome**: IQ ~50 but fluent language. **Evidence**: vocal tract adaptation · Broca's/Wernicke's areas · creole formation (Bickerton) · Nicaraguan signing creole (Pinker, 1994). **Limitations**: theoretical only · no account of interaction · **Jim case** (Bard & Sachs, 1977): exposure without interaction insufficient.

Critical Period Biology

Brain: decline in metabolic rate + neuron numbers during early school years; synapses/metabolic rate bottom out around puberty. Children recover language after left-hemisphere damage; adults → permanent aphasia (Curtiss, 1989; Lenneberg, 1967).

Piaget's Cognitive Stages

Sensorimotor (0–2; lacks object permanence) → **Preoperational** (2–7; language develops; egocentric) → **Concrete operational** (7–12; conservation; reversibility) → **Formal operational** (12+; abstract; only 40–60% adults). Brown (1973): morpheme mastery order = -ing → in/on → plural -s → forms of "to be."

Unit 2: Language Processing

Clark & Clark (1977) — 3 Elements of Expression

Speech Acts — Searle (1969); indirect speech acts; context-dependent meaning (Gibbs, 1986). **Propositional Content** — describes states/events; more propositions = longer reading time (van Dijk & Kintsch, 1983). **Thematic Structure** — good speakers judge what listeners know.

5 Stages of Speech Production (Clark & Clark, 1977)

Discourse plan → **Sentence plan** → **Constituent plan** → **Articulatory program** → **Articulation**

Gricean Maxims (Grice, 1975)

Quantity (avoid running off) · **Quality** (don't lie) · **Relation** (avoid irrelevance) · **Manner** (avoid ambiguity). Violation = **conversational implicature**.

Deep vs. Surface Structure (Chomsky, 1965, 1975)

Transformational rules. Deep structure = underlying meaning. Surface structure = actual spoken form. Synonymous sentences = same deep, different surface. Ambiguous = different deep, same surface.

Speech Errors

"Slips seldom illegal combinations; morphemes slip as entire units." **Bloopers**: "homely friendmaker" for "Friendly Homemaker Program."

Unit 3: Multilingualism & Cognition

Definitions

Bloomfield (1933): native-like competence in all. Modern (Grosjean, 1982): uses languages regularly, switches where necessary. **Elite** (formal setting) vs **Neighbourhood** (natural interaction). **Simultaneous** vs **Sequential** bilingualism.

Additive vs Subtractive (Cummins, 1976)

Additive: L2 added to well-developed L1 → positive cognitive effects. **Subtractive**: L2 replaces L1 → negative effects. **Threshold effect**: high competence in both needed.

Cognitive Benefits

Enhanced metalingual abilities (Bialystok, 2001) · Better problem solving (Cummins, 1981) · Better verbal+spatial (Diaz, 1983) · Greater cognitive flexibility (Hakuta, 1986) · Better creativity/analogical reasoning (Diaz, 1985) · Divergent thinking (Landry, 1968–74) · Competent multilinguals highest on analogy-solving (Eleanor, 1993).

Code-Switching

Matrix language (scaffolding) + embedded language (fills gaps). **Myers Scotton (1993)**: embedded language island. **MacSwan (1997, 1999)**: nothing constrains code switching apart from mixed grammars.

Single vs. Dual System

Single: one brain region (Hernandez et al., 2001). **Dual**: separate systems (De Houwer, 1995; Paradis, 1981). Bilingual aphasia evidence equivocal. **Ojemann & Whitaker (1978)**: cortical stimulation → "some singly, others separately." Early bilinguals = adaptable network; late = new neural areas.

Second Language Acquisition

No critical period for L2 except native accent (Birdsong, 1999). L2 = cognitive procedures + LAD (less for L2) + LASS + Vygotskyan scaffolding. Differential difficulty depends on shared roots/inflection.

Unit 4: Language & Speech Disorders

Aphasia

Broca's: premotor area; non-fluent, agrammatic; comprehension preserved. **Wernicke's**: left temporal lobe; fluent but semantically incoherent; **neologisms**. **Global**: both areas; 32% post-stroke. **Anomic**: word-retrieval difficulty. **Speech Disorders (Roseberry-McKibbin, 1995)**

Voice (phonation/resonation) · **Articulation** (phonetic vs phonemic) · **Fluency** (cluttering, stuttering) · **Apraxia** (acquired vs developmental DAS) · **Dysprosody** (rhythm/intonation) · **Dysarthria** (muscle weakness/paralysis).

Other Disorders

Autism (Kanner, 1943): echolalia; ~50% don't develop functional speech. **Alzheimer's**: semantic simplification; phonology/syntax minimally affected. **SLI**: deficit in function morphemes; ~50% → dyslexia (Bishop & Snowling, 2004). **Dyspraxia** (Dewey, 1995): impairment planning/executing motor tasks; lifelong; "out of sync." DAS: developmental apraxia of speech. **Dyslexia** (WFN, 1968): reading/writing/spelling disorder; primary (hereditary) vs secondary (hormonal). Subtypes: visual, auditory, dysgraphia.